Лабораторная работа №10

Расширенные настройки SMTP-сервера

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Информация

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Цели и задачи

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- Приобретение практических навыков по конфигурированию SMTP-сервера в части настройки аутентификации.

1. Настройте Dovecot для работы с LMTP (см. раздел 10.4.1).
2. Настройте аутентификацию посредством SASL на SMTP-сервере (см. раздел 10.4.2).
3. Настройте работу SMTP-сервера поверх TLS (см. раздел 10.4.3).
4. Скорректируйте скрипт для Vagrant, фиксирующий действия расширенной настройки SMTP-сервера во внутреннем окружении виртуальной машины server (см. раздел 10.4.4).

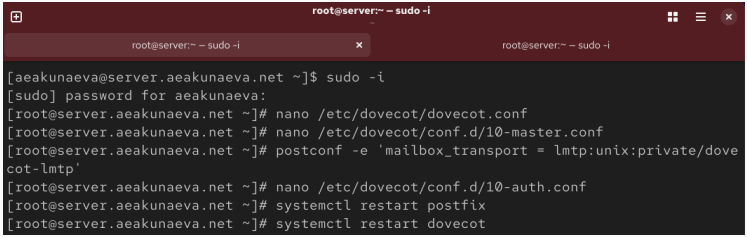
Материалы и методы

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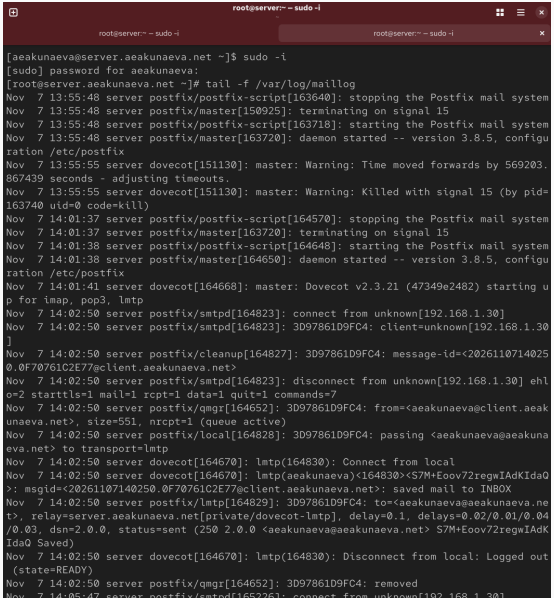
- Linux (дистрибутив Rocky 10.2)
- Linux Fedora Workstation (Markdown)
- Oracle VirtualBox 7.2.16
- Vagrant 2.4.9 (Windows amd64)
- FAR Manager 3.0 build 6699 x64
- Packer 1.16.0 (Windows amd64)
- Postfix
- Dovecot
- Telnet

Выполнение лабораторной работы

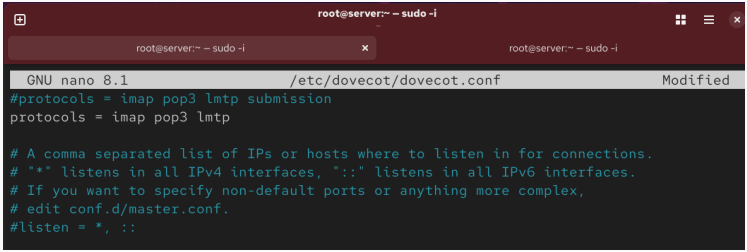
Настройка LMTP и Dovecot



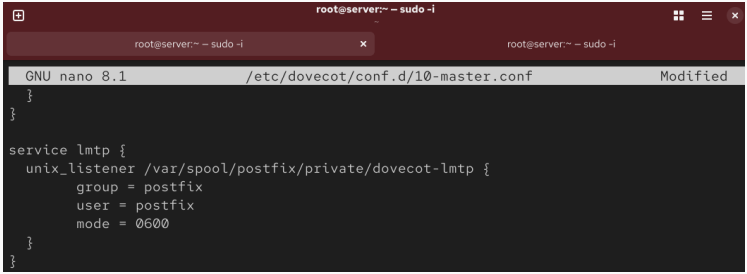
Журнал мониторинга работы почтовой службы



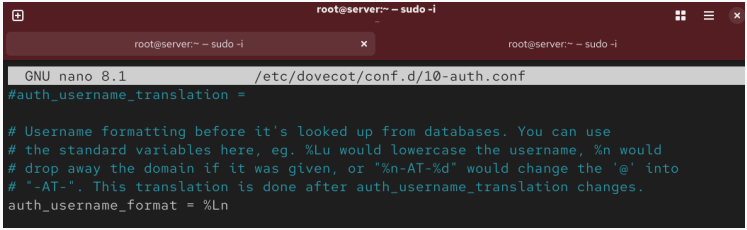
Редактирование /etc/dovecot/dovecot.conf



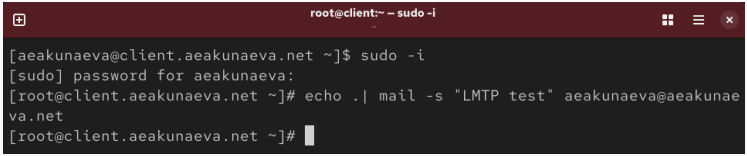
Редактирование /etc/dovecot/conf.d/10-master.conf



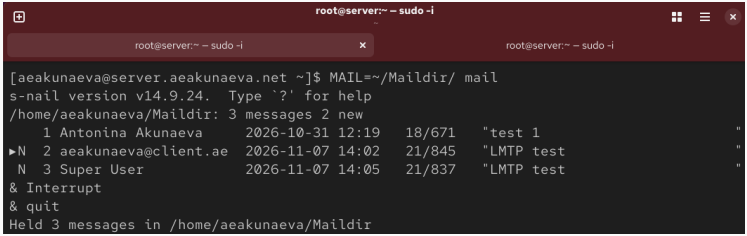
Редактирование etc/dovecot/conf.d/10-auth.conf



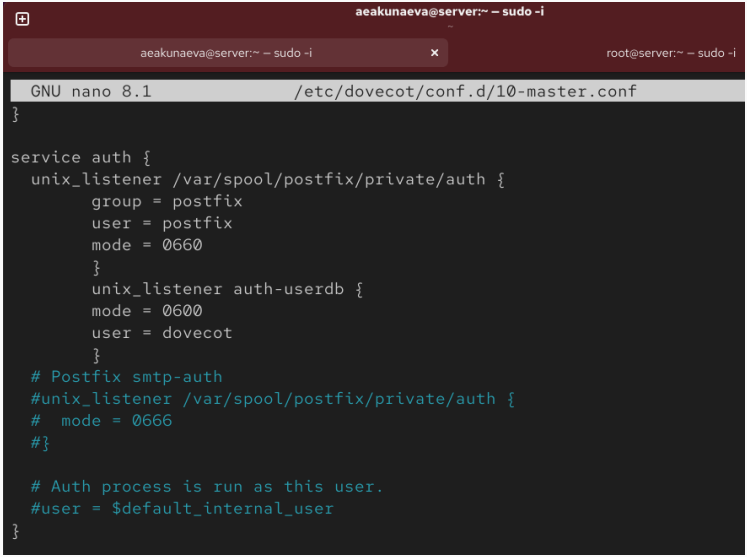
Отправление письма с client через LMTP



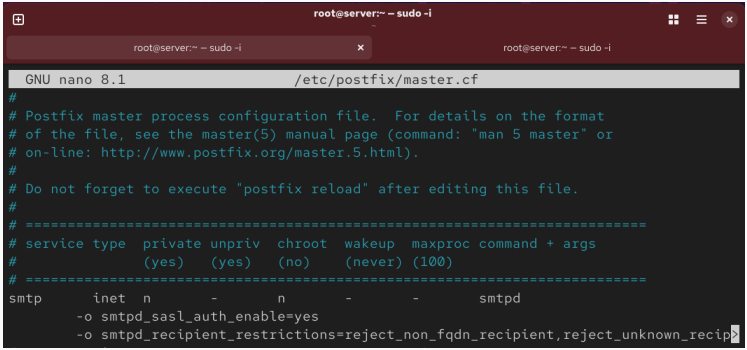
Проверка почтового ящика на server



Редактирование /etc/dovecot/conf.d/10-master.conf



Редактирование /etc/postfix/master.cf



Настройка Postfix

```
[aeakunaeva@server.aeakunaeva.net ~]$ nano /etc/dovecot/conf.d/10-master.conf
[aeakunaeva@server.aeakunaeva.net ~]$ sudo -i
[sudo] password for aeakunaeva:
[root@server.aeakunaeva.net ~]# nano /etc/dovecot/conf.d/10-master.conf
[root@server.aeakunaeva.net ~]# postconf -e 'smtpd_sasl_type = dovecot
> '
postconf: fatal: -e, -X, or -# accepts no multi-line input
[root@server.aeakunaeva.net ~]# postconf -e 'smtpd_sasl_type = dovecot'
[root@server.aeakunaeva.net ~]# postconf -e 'smtpd_sasl_path = private/auth'
[root@server.aeakunaeva.net ~]# postconf -e 'smtpd_recipient_restrictions =
reject_unknown_recipient_domain, permit_mynetworks,
reject_non_fqdn_recipient, reject_unauth_destination,
reject_unverified_recipient, permit'
postconf: fatal: -e, -X, or -# accepts no multi-line input
[root@server.aeakunaeva.net ~]# postconf -e 'smtpd_recipient_restrictions = reject_unkn
own_recipient_domain, permit_mynetworks, reject_non_fqdn_recipient, reject_unauth_desti
nation, reject_unverified_recipient, permit'
[root@server.aeakunaeva.net ~]# postconf -e 'mynetworks = 127.0.0.0/8'
[root@server.aeakunaeva.net ~]# nano /etc/postfix/master.cf
[root@server.aeakunaeva.net ~]# systemctl restart postfix
[root@server.aeakunaeva.net ~]# systemctl restart dovecot
```

Установка telnet на client

```
[root@client.aeakunaeva.net ~]# dnf -y install telnet
Extra Packages for Enterprise Linux 10 - x86_6  29 kB/s |  32 kB     00:01
Extra Packages for Enterprise Linux 10 - x86_6 4.3 MB/s | 6.7 MB     00:01
Rocky Linux 10 - BaseOS                        6.0 kB/s | 4.3 kB     00:00
Rocky Linux 10 - BaseOS                         25 MB/s |  45 MB     00:01
Rocky Linux 10 - AppStream                      13 kB/s | 4.4 kB     00:00
Rocky Linux 10 - AppStream                     4.1 MB/s | 2.9 MB     00:00
Rocky Linux 10 - CRB                           8.7 kB/s | 4.3 kB     00:00
Rocky Linux 10 - CRB                           616 kB/s | 570 kB     00:00
Rocky Linux 10 - Extras                        9.0 kB/s | 3.1 kB     00:00
Dependencies resolved.
================================================================================
 Package        Architecture   Version                 Repository         Size
================================================================================
Installing:
 telnet         x86_64         1:0.17-94.el10          appstream          62 k

Transaction Summary
================================================================================
Install  1 Package

Total download size: 62 k
Installed size: 109 k
Downloading Packages:
telnet-0.17-94.el10.x86_64.rpm                 173 kB/s |  62 kB     00:00
--------------------------------------------------------------------------------
Total                                          102 kB/s |  62 kB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
```

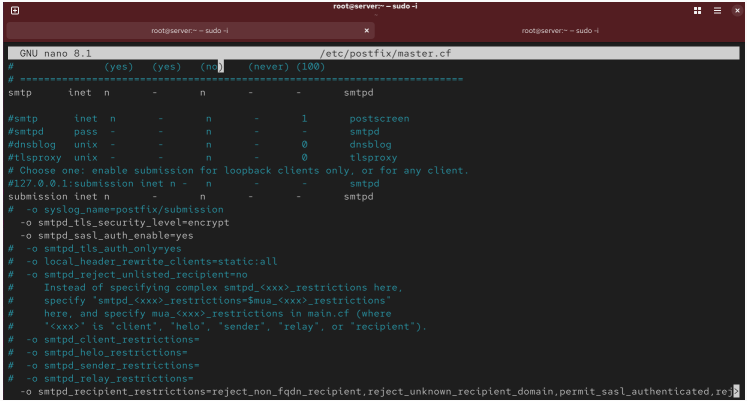
Проверка работы telnet на client

```
[root@client.aeakunaeva.net ~]# printf 'aeakunaeva\x00aeakunaeva\x00123456' | b
ase64
YWVha3VuYWV2YQBhZWFrdW5hZXZhADEyMzQ1Ng==
[root@client.aeakunaeva.net ~]# telnet server.aeakunaeva.net 25
Trying 192.168.1.1...
Connected to server.aeakunaeva.net.
Escape character is '^]'.
220 server.aeakunaeva.net ESMTP Postfix
EHLO test
250-server.aeakunaeva.net
250-PIPELINING
250-SIZE 10240000
250-VRFY
250-ETRN
250-STARTTLS
250-AUTH PLAIN
250-ENHANCEDSTATUSCODES
250-8BITMIME
250-DSN
250-SMTPUTF8
250 CHUNKING
AUTH PLAIN YWVha3VuYWV2YQBhZWFrdW5hZXZhADEyMzQ1Ng==
235 2.7.0 Authentication successful
quit
221 2.0.0 Bye
Connection closed by foreign host.
[root@client.aeakunaeva.net ~]#
```

Конфигурация Postfix

```
[root@server.aeakunaeva.net ~]# cp /etc/pki/dovecot/certs/dovecot.pem /etc/pki/tls/certs
[root@server.aeakunaeva.net ~]# cp /etc/pki/dovecot/private/dovecot.pem /etc/pki/tls/private
[root@server.aeakunaeva.net ~]# postconf -e 'smtpd_tls_cert_file=/etc/pki/tls/certs/dovecot.pem'
[root@server.aeakunaeva.net ~]# postconf -e 'smtpd_tls_key_file=/etc/pki/tls/private/dovecot.pem'
[root@server.aeakunaeva.net ~]# postconf -e 'smtpd_tls_session_cache_database = btree:/var/lib/postfix/smtpd_scache'
[root@server.aeakunaeva.net ~]# postconf -e 'smtpd_tls_security_level = may'
[root@server.aeakunaeva.net ~]# postconf -e 'smtp_tls_security_level = may'
[root@server.aeakunaeva.net ~]# nano /etc/postfix/master.cf
[root@server.aeakunaeva.net ~]# firewall-cmd --get-services
0-AD RH-Satellite-6 RH-Satellite-6-capsule afp alvr amanda-client amanda-k5-client amqp amqps anno-1602 anno-1800 apcupsd aseqnet audit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent civilization-iv civilization-v cockpit collectd condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-quic dns-over-tls docker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server factorio finger foreman foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-client ganglia-master git gitea gpsd grafana gre high-availability http http3 https ident imap imaps iperf2 iperf3 ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-manager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp managesieve m
```

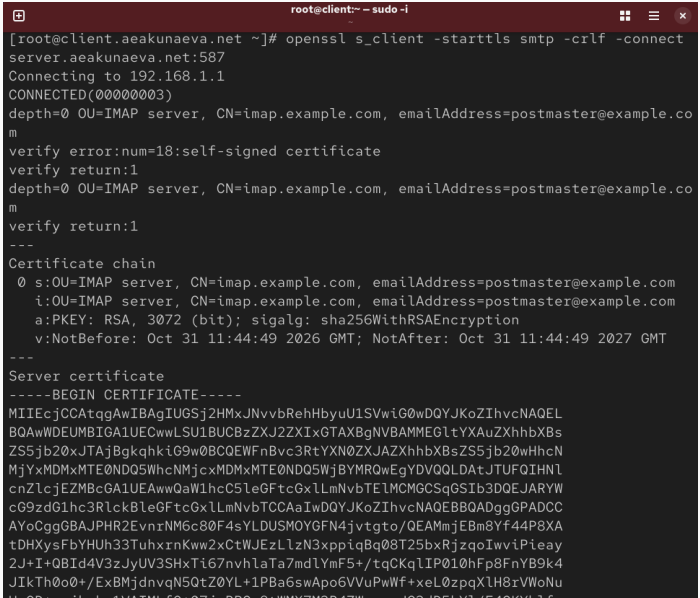
Редактирование /etc/postfix/master.cf



Настройка межсетевого экрана

```
sh rsyncd rtsp salt-master samba samba-client samba-dc sane settlers-history-collection
 sip sips slimevr slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptrap soc
ks spideroak-lansync spotify-sync squid ssdp ssh ssh-custom statsrv steam-lan-transfer
steam-streaming stellaris stronghold-crusader stun stuns submission supertuxkart svdrp
svn syncthing syncthing-gui syncthing-relay synergy syscomlan syslog syslog-ng syslog-t
ls telnet tentacle terraria tftp tile38 tinc tor-socks transmission-client turn turns u
pnp-client vdsm vnc-server vrrp warpinator wbem-http wbem-https wireguard ws-discovery
ws-discovery-client ws-discovery-host ws-discovery-tcp ws-discovery-udp wsdd wsdd-http
wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-jav
a-gateway zabbix-server zabbix-trapper zabbix-web-service zero-k zerotier
[root@server.aeakunaeva.net ~]# firewall-cmd --add-service=smtp-submission
success
[root@server.aeakunaeva.net ~]# firewall-cmd --add-service=smtp-submission --permanent
success
[root@server.aeakunaeva.net ~]# firewall-cmd --reload
success
[root@server.aeakunaeva.net ~]# systemctl restart postfix
[root@server.aeakunaeva.net ~]#
```

Подключение к SMTP-серверу на client

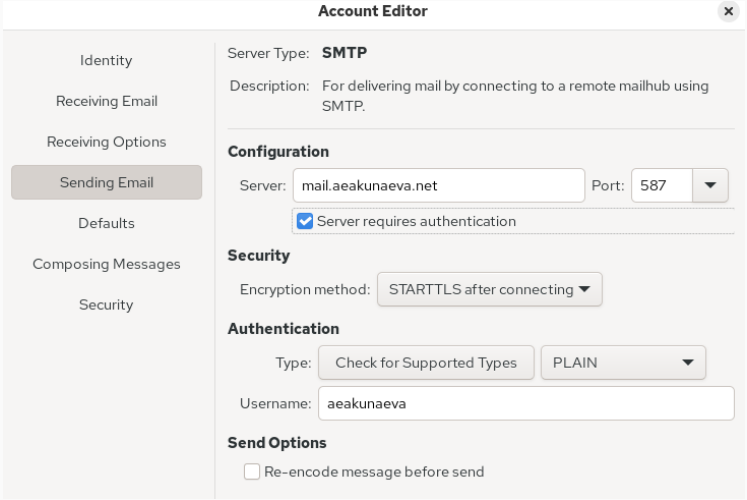


Подключение к SMTP-серверу на client

```
    00b0 - 3a 3d 88 cb 9a 65 63 c6-cf 01 66 77 e3 d1 4b e3   .=...ec...fw..K.
    00c0 - d0 34 70 49 79 82 b4 f4-1c b1 f3 92 ca 6f e0 59   .4pIy........o.Y

    Start Time: 1794062141
    Timeout   : 7200 (sec)
    Verify return code: 18 (self-signed certificate)
    Extended master secret: no
    Max Early Data: 0
---
read R BLOCK
EHLO test
250-server.aeakunaeva.net
250-PIPELINING
250-SIZE 10240000
250-VRFY
250-ETRN
250-AUTH PLAIN
250-ENHANCEDSTATUSCODES
250-8BITMIME
250-DSN
250-SMTPUTF8
250 CHUNKING
AUTH PLAIN YWVha3VuYWV2YQBhZWFrdW5hZXZhADEyMzQ1Ng==
235 2.7.0 Authentication successful
quit
221 2.0.0 Bye
closed
[root@client.aeakunaeva.net ~]# evolution
```

Настройка учётной записи Evolution



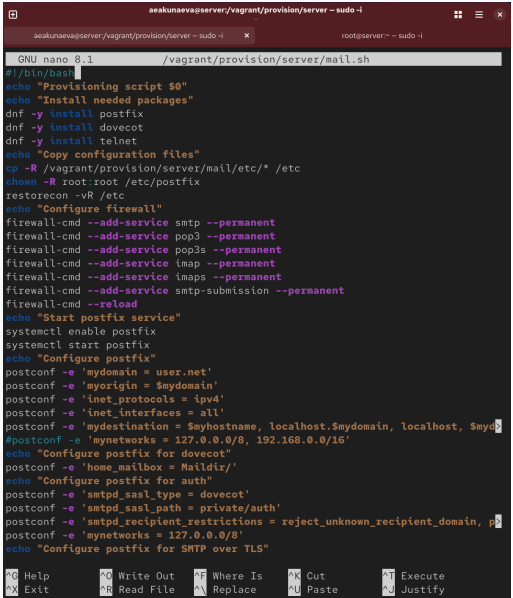
Проверка корректности отправки почтовых сообщений

```
[root@server.aeakunaeva.net ~]# su - aeakunaeva
Last login: Sat Nov  7 14:06:37 UTC 2026 on pts/0
[aeakunaeva@server.aeakunaeva.net ~]$ MAIL=~/Maildir/ mail
s-nail version v14.9.24.  Type `?' for help
/home/aeakunaeva/Maildir: 4 messages 1 new 1 deleted
    1 Antonina Akunaeva     2026-10-31 12:19   18/671   "test 1
    2 aeakunaeva@client.ae  2026-11-07 14:02   21/845   "LMTP test
▸N  4 Antonina Akunaeva     2026-11-07 15:29   22/839   "test smtp
& Interrupt
&
```

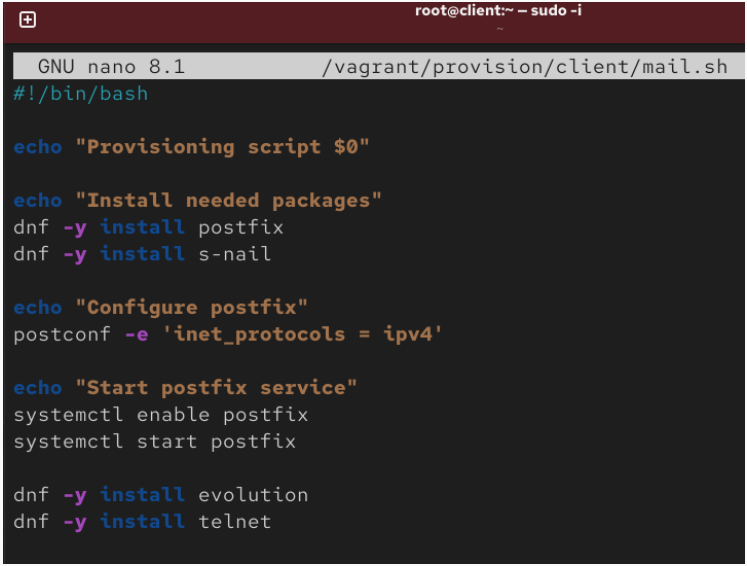
Внесение изменений в настройки внутреннего окружения server

```
[aeakunaeva@server.aeakunaeva.net ~]$ cd /vagrant/provision/server
[aeakunaeva@server.aeakunaeva.net server]$ cd /vagrant/provision/server /vagran
t/provision/server/mail/etc/dovecot/
-bash: cd: too many arguments
[aeakunaeva@server.aeakunaeva.net server]$ cp -R /etc/dovecot/dovecot.conf /vag
rant/provision/server/mail/etc/dovecot/
[aeakunaeva@server.aeakunaeva.net server]$ cp -R /etc/dovecot/conf.d/10-master.
conf /vagrant/provision/server/mail/etc/dovecot/conf.d/
[aeakunaeva@server.aeakunaeva.net server]$ cp -R /etc/dovecot/conf.d/10-auth.co
nf /vagrant/provision/server/mail/etc/dovecot/conf.d/
[aeakunaeva@server.aeakunaeva.net server]$ mkdir -p /vagrant/provision/server/m
ail/etc/postfix/
[aeakunaeva@server.aeakunaeva.net server]$ cp -R /etc/postfix/master.cf /vagran
t/provision/server/mail/etc/postfix/
[aeakunaeva@server.aeakunaeva.net server]$ nano /vagrant/provision/server/mail.
sh
[aeakunaeva@server.aeakunaeva.net server]$
```

Редактирование /vagrant/provision/server/mail.sh



Редактирование /vagrant/provision/client/mail.sh



Выводы

Выводы

Я приобрела практические навыки по конфигурированию SMTP-сервера в части настройки аутентификации.